PHẦN TỬ DSG3

LÝ THUYẾT TẤM REISSNER\MINDLIN

Trình bày: Phùng Văn Phúc

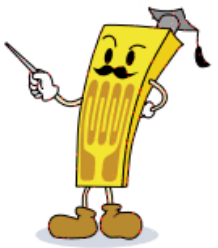
GVHD: TS. Nguyễn Thời Trung

Tp. HCM 21-07-2011

Nội dung trình bày

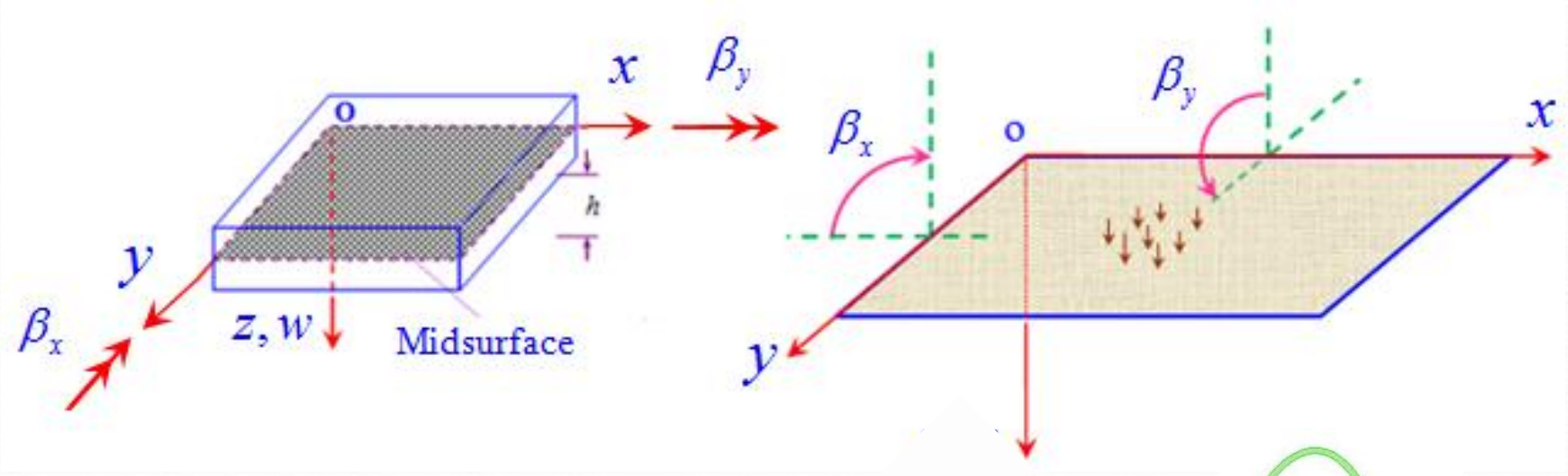


- Quy ước dấu trong tấm Reissner\Mindlin
- Cách thành lập Bb (biến dạng uốn)
- Xử lý ma trận Bs (biến dạng cắt)
- Phần tử MIN3
- Phần tử DKT



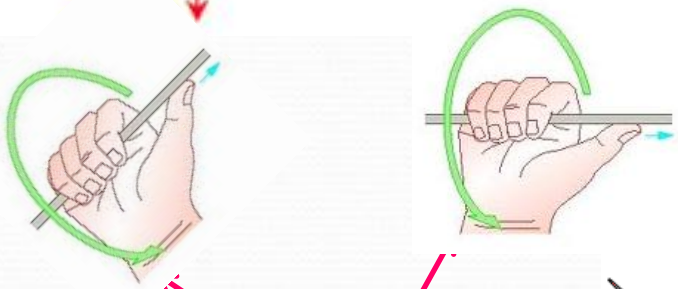
Phần tử DSG3

➢ Quy ước dấu trong tấm Reissner\Mindlin



- β_y là góc quay xung quanh trục x
- β_x là góc quay xung quanh trục y

➔ Xác định theo quy tắc bàn tay phải

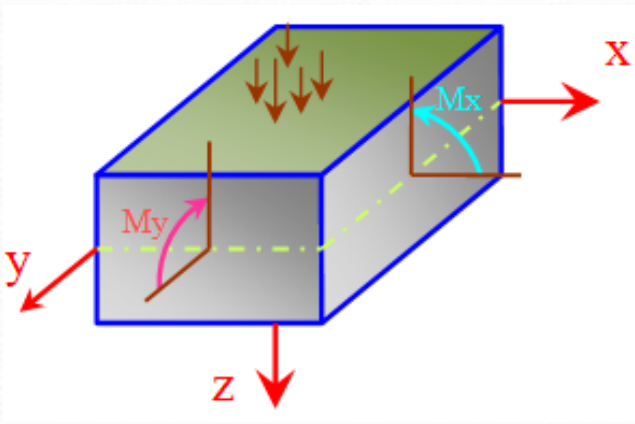


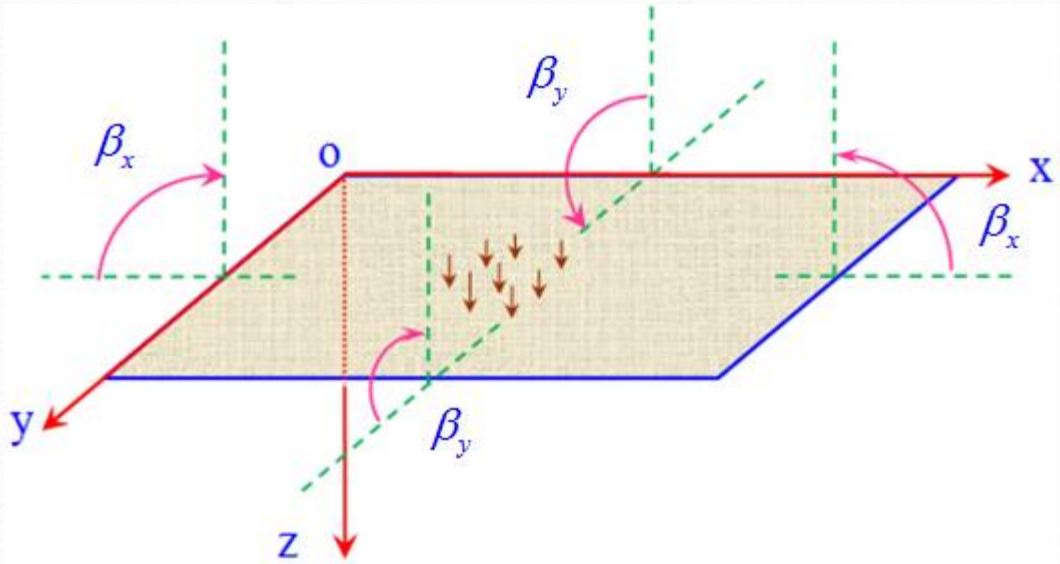
Chiều quay của β theo độ võng

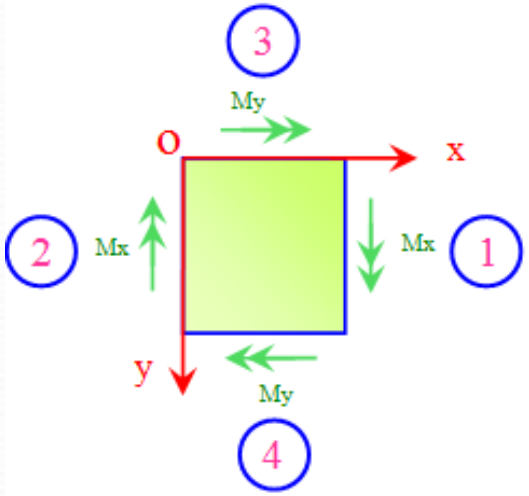
Phần tử DSG3



➢ Quy ước dấu trong tấm Reissner\Mindlin





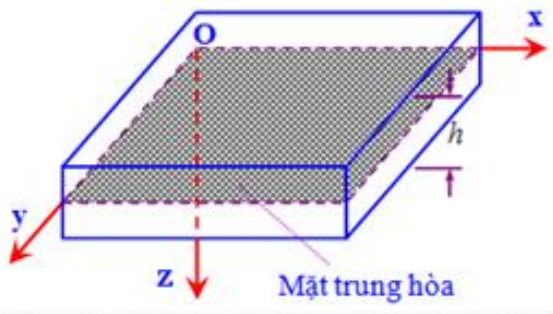


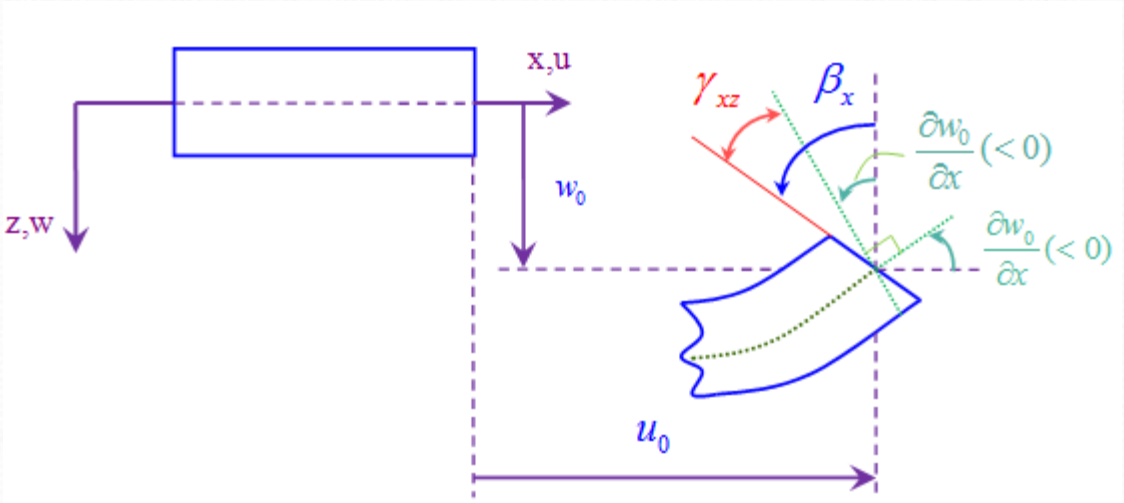
○ Chiều quay của β chính là chiều của moment



Phần tử DSG3

➢ Quy ước dấu trong tấm Reissner\Mindlin





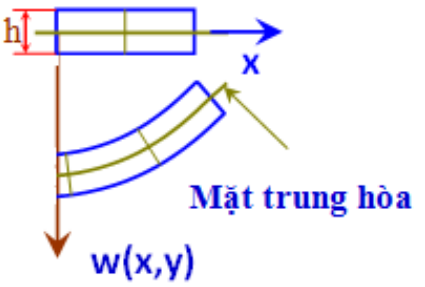
$\frac{\partial w_0}{\partial x}(<0)$ Góc xoay do biến dạng uốn

γ_{xz} Góc xoay do biến dạng cắt

β_x Góc xoay tổng

Theo trục x,

w giảm

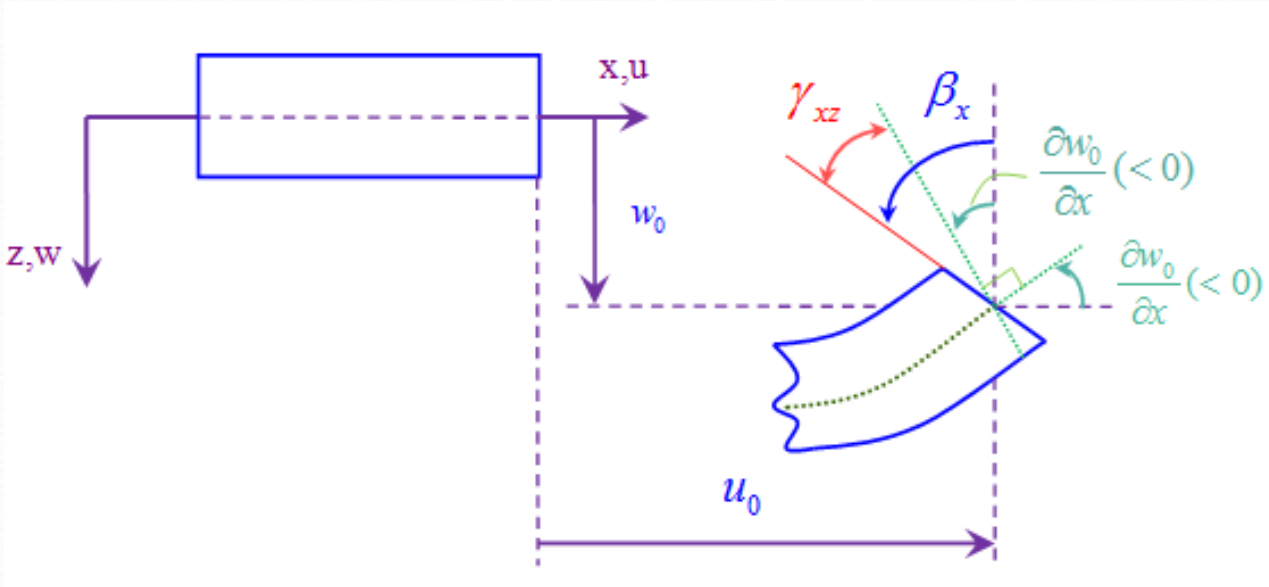


$$\frac{\partial w_o}{\partial x} = \lim_{\Delta x \to 0} \frac{w_o(x+\Delta x) - w_0(x)}{\Delta x} < 0$$

Phần tử DSG3



➢ Quy ước dấu trong tấm Reissner\Mindlin

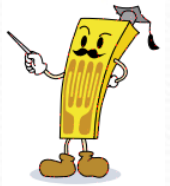


○ Góc xoay do biến dạng cắt gây ra

$$\gamma_{xz} = \frac{\partial w_0}{\partial x} + \beta_x = w'_x + \beta_x$$

○ Chuyển vị theo phương x

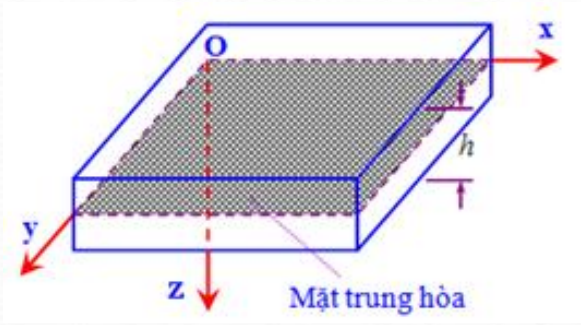
$$u = u_0 + z\beta_x$$

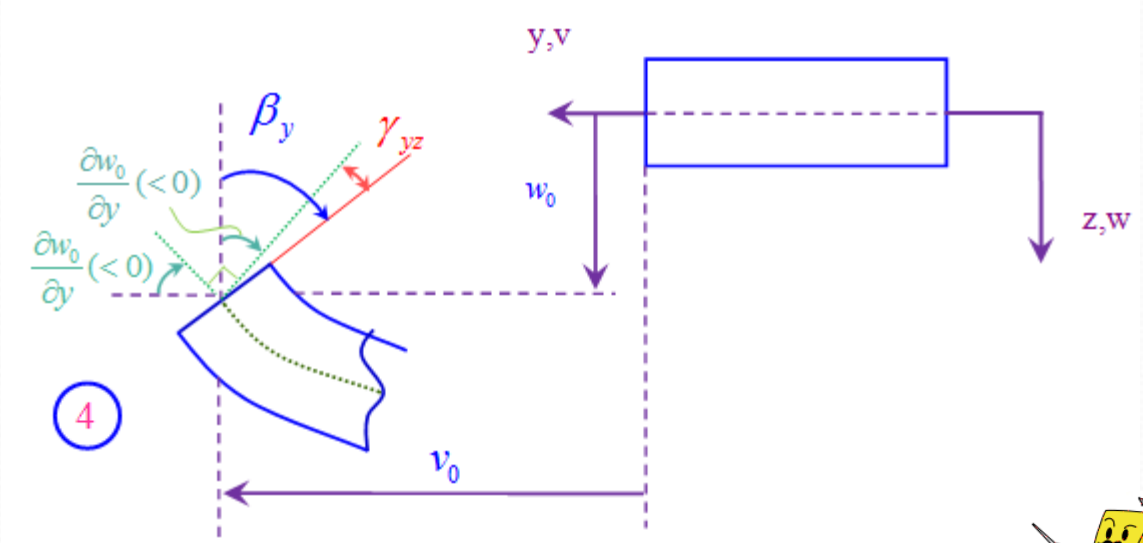


Phần tử DSG3



➢ Quy ước dấu trong tấm Reissner\Mindlin







$\frac{\partial w_0}{\partial y}(<0)$ Góc xoay do biến dạng uốn

γ_{yz} Góc xoay do biến dạng cắt

β_y Góc xoay tổng

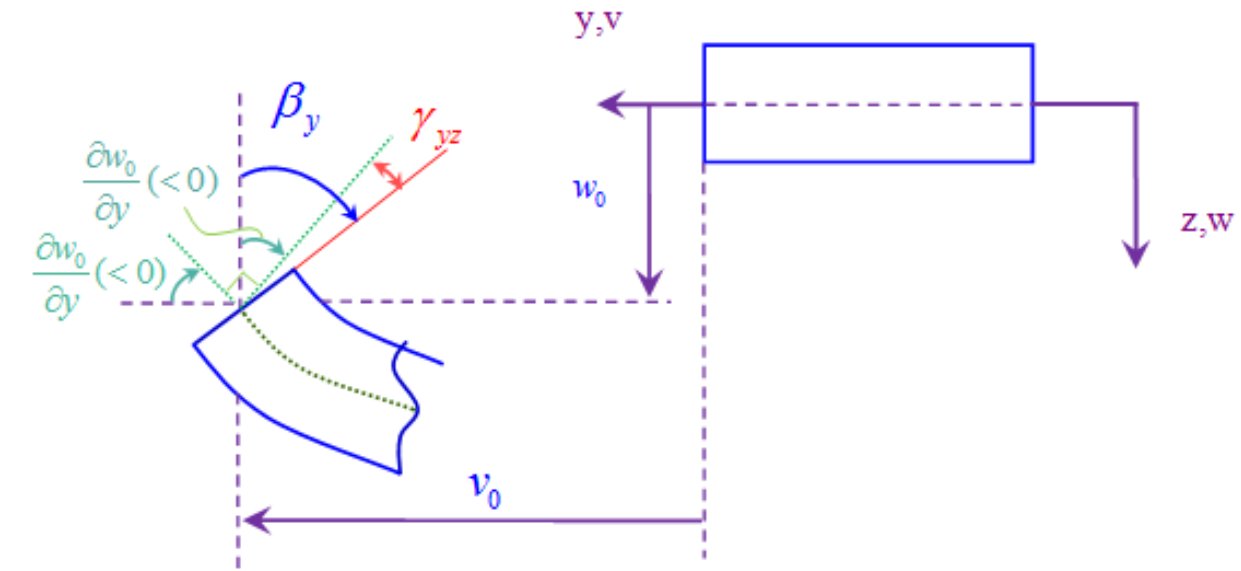
Theo trục y, *w* giảm

$$\frac{\partial w_o}{\partial y} = \lim_{\Delta y \to 0} \frac{w_o(y+\Delta y) - w_0(y)}{\Delta y} < 0$$

Phần tử DSG3



➢ Quy ước dấu trong tấm Reissner\Mindlin



○ Góc xoay do biến dạng cắt gây ra

$$\gamma_{yz} = \frac{\partial w_0}{\partial y} + \beta_y = w'_y + \beta_y$$

○ Chuyển vị theo phương x

$$v = v_0 + z\beta_y$$

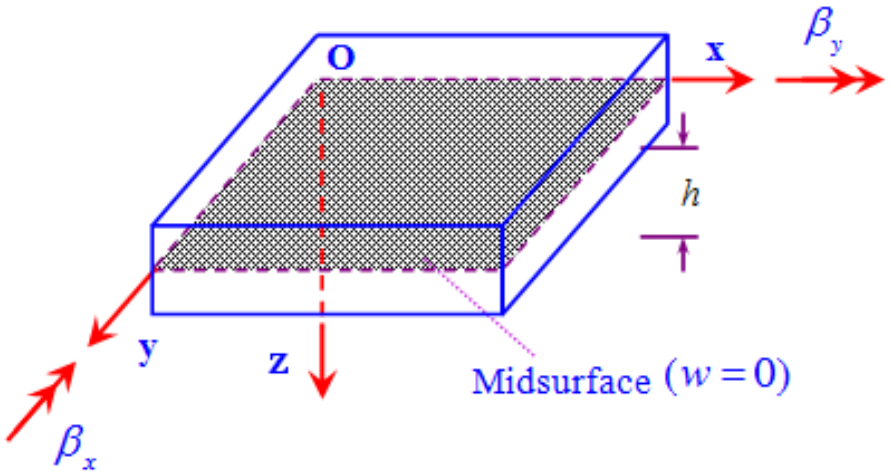


Phần tử DSG3

➢ Thành lập công thức Bb

○ Chuyển vị theo 3 phương x – y - z

$$u = u_0 + z\beta_x$$
$$v = v_0 + z\beta_y$$
$$w = w_0$$



○ Biến xấp xỉ theo thứ tự z – y - x

$$\mathbf{U}_\beta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}$$

$w_i \Rightarrow$ Trục z

$\beta_{xi} \Rightarrow$ Trục y

$\beta_{yi} \Rightarrow$ Trục x

$$\mathbf{U}_\beta = \mathbf{N}_\beta \mathbf{d}_\beta$$

Phần tử DSG3

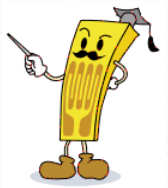


➢ Thành lập công thức Bb

○ Biến dạng uốn $u = u_0 + z\beta_x \quad v = v_0 + z\beta_y$

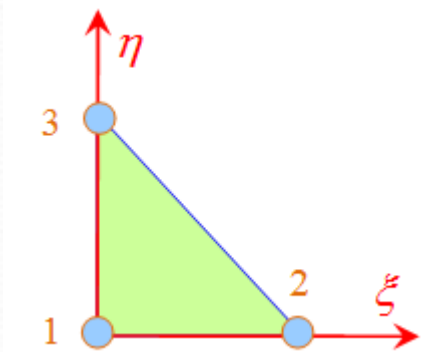
$$\varepsilon_b = \begin{bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial}{\partial x}(u_0 + z\beta_x) \\ \dfrac{\partial v}{\partial y}(v_0 + z\beta_y) \\ \dfrac{\partial}{\partial y}(u_0 + z\beta_x) + \dfrac{\partial}{\partial x}(v_0 + z\beta_y) \end{bmatrix} = z \begin{bmatrix} \dfrac{\partial \beta_x}{\partial x} \\ \dfrac{\partial \beta_y}{\partial y} \\ \dfrac{\partial \beta_x}{\partial y} + \dfrac{\partial \beta_y}{\partial x} \end{bmatrix} = z\kappa$$

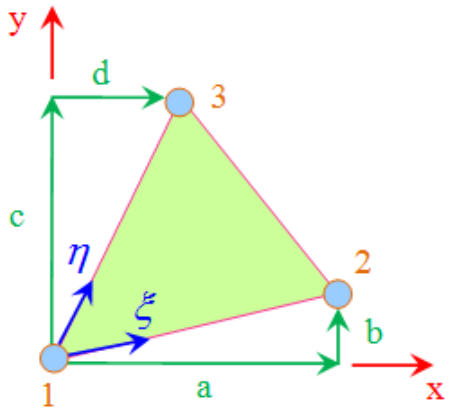
$$\kappa = \underbrace{\begin{bmatrix} 0 & \dfrac{\partial}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial}{\partial y} \\ 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix}}_{\mathbf{U}_\beta} = \begin{bmatrix} 0 & \dfrac{\partial}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial}{\partial y} \\ 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \underbrace{\begin{bmatrix} 0 & \dfrac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial N_i}{\partial y} \\ 0 & \dfrac{\partial N_i}{\partial y} & \dfrac{\partial N_i}{\partial x} \end{bmatrix}}_{\mathbf{B}^b} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^b \mathbf{d}_\beta$$



Phần tử DSG3

➢ Thành lập công thức Bb





$$a = x_2 - x_1$$
$$b = y_2 - y_1$$
$$c = y_3 - y_1$$
$$d = x_3 - x_1$$

$$\xi_1 = 0 \quad \eta_1 = 0 \quad N_1 = 1 - \xi - \eta$$
$$\xi_2 = 1 \quad \eta_1 = 0 \quad N_2 = \xi$$
$$\xi_1 = 0 \quad \eta_1 = 1 \quad N_3 = \eta$$

$$N_{1,\xi} = -1 \quad N_{2,\xi} = 1 \quad N_{3,\xi} = 0$$
$$N_{1,\eta} = -1 \quad N_{2,\eta} = 0 \quad N_{3,\eta} = 1$$

$$x_{,\xi} = N_{1,\xi}x_1 + N_{2,\xi}x_2 + N_{3,\xi}x_3 = x_2 - x_1 = a$$
$$y_{,\xi} = N_{1,\xi}y_1 + N_{2,\xi}y_2 + N_{3,\xi}y_3 = y_2 - y_1 = b$$
$$x_{,\eta} = N_{1,\eta}x_1 + N_{2,\eta}x_2 + N_{3,\eta}x_3 = x_3 - x_1 = d$$
$$y_{,\eta} = N_{1,\eta}y_1 + N_{2,\eta}y_2 + N_{3,\eta}y_3 = y_3 - y_1 = c$$

$$\mathbf{J} = \begin{bmatrix} x_{,\xi} & y_{,\xi} \\ x_{,\eta} & y_{,\eta} \end{bmatrix} = \begin{bmatrix} a & b \\ d & c \end{bmatrix} \Rightarrow \mathbf{J}^{-1} = \begin{bmatrix} \xi_{,x} & \eta_{,x} \\ \xi_{,y} & \eta_{,y} \end{bmatrix} = \frac{1}{\det \mathbf{J}} \begin{bmatrix} c & -b \\ -d & a \end{bmatrix}$$

Phần tử DSG3

➢ Thành lập công thức Bb

$$\frac{\partial N_i}{\partial x}=\frac{\partial N_i}{\partial \xi}\frac{\partial \xi}{\partial x}+\frac{\partial N_i}{\partial \eta}\frac{\partial \eta}{\partial x}=N_{i,\xi}\xi_{,x}+N_{i,\eta}\eta_{,x}$$

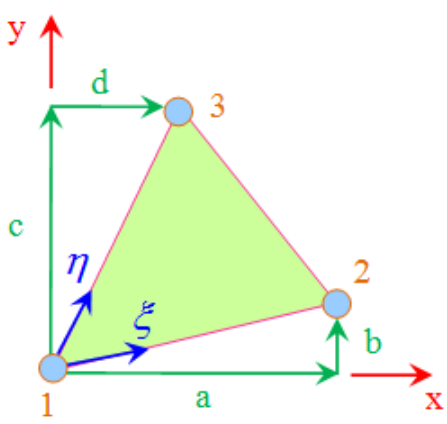
$$\frac{\partial N_1}{\partial x}=N_{1,\xi}\xi_{,x}+N_{1,\eta}\eta_{,x}=\underbrace{(-1)}_{N_{1,\xi}}\frac{\mathrm{c}}{\det \mathbf{J}}+\underbrace{(-1)}_{N_{1,\eta}}\frac{-\mathrm{b}}{\det \mathbf{J}}=\frac{\mathrm{b}-\mathrm{c}}{\det \mathbf{J}}$$

$$\frac{\partial N_2}{\partial x}=\frac{\mathrm{c}}{\det \mathbf{J}};\quad \frac{\partial N_3}{\partial x}=\frac{-\mathrm{b}}{\det \mathbf{J}}$$

$$\frac{\partial N_i}{\partial y}=\frac{\partial N_i}{\partial \xi}\frac{\partial \xi}{\partial y}+\frac{\partial N_i}{\partial \eta}\frac{\partial \eta}{\partial y}=N_{i,\xi}\xi_{,y}+N_{i,\eta}\eta_{,y}$$

$$\frac{\partial N_1}{\partial y}=N_{1,\xi}\xi_{,y}+N_{1,\eta}\eta_{,y}=\underbrace{(-1)}_{N_{1,\xi}}\frac{-\mathrm{d}}{\det \mathbf{J}}+\underbrace{(-1)}_{N_{1,\eta}}\frac{\mathrm{a}}{\det \mathbf{J}}=\frac{\mathrm{d}-\mathrm{a}}{\det \mathbf{J}}$$

$$\frac{\partial N_2}{\partial y}=\frac{-\mathrm{d}}{\det \mathbf{J}};\quad \frac{\partial N_3}{\partial y}=\frac{\mathrm{a}}{\det \mathbf{J}}$$



$$N_1=1-\xi-\eta \quad N_2=\xi \quad N_3=\eta$$

$$N_{1,\xi}=-1 \quad N_{2,\xi}=1 \quad N_{3,\xi}=0$$

$$N_{1,\eta}=-1 \quad N_{2,\eta}=0 \quad N_{3,\eta}=1$$

$$\mathbf{J}^{-1}=\begin{bmatrix}\xi_{,x} & \eta_{,x}\\ \xi_{,y} & \eta_{,y}\end{bmatrix}=\frac{1}{\det \mathbf{J}}\begin{bmatrix}\mathrm{c} & -\mathrm{b}\\ -\mathrm{d} & \mathrm{a}\end{bmatrix}$$

Phần tử DSG3



➢ Thành lập công thức Bb

$$\frac{\partial N_1}{\partial x}=\frac{b-c}{\det \mathbf{J}};\quad \frac{\partial N_2}{\partial x}=\frac{c}{\det \mathbf{J}};\quad \frac{\partial N_3}{\partial x}=\frac{-b}{\det \mathbf{J}}$$

$$\frac{\partial N_1}{\partial y}=\frac{d-a}{\det \mathbf{J}};\quad \frac{\partial N_2}{\partial y}=\frac{-d}{\det \mathbf{J}};\quad \frac{\partial N_3}{\partial y}=\frac{a}{\det \mathbf{J}}$$

$$\mathbf{B}^b=\begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial y} \end{bmatrix}=\left[\begin{bmatrix} 0 & \frac{\partial N_1}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial y} \\ 0 & \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial y} \end{bmatrix} \begin{bmatrix} 0 & \frac{\partial N_2}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_2}{\partial y} \\ 0 & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial y} \end{bmatrix} \begin{bmatrix} 0 & \frac{\partial N_3}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_3}{\partial y} \\ 0 & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial y} \end{bmatrix}\right]$$

$$\mathbf{B}^b=\frac{1}{\det \mathbf{J}}\left[\begin{bmatrix} 0 & b-c & 0 \\ 0 & 0 & d-a \\ 0 & d-a & b-c \end{bmatrix} \begin{bmatrix} 0 & c & 0 \\ 0 & 0 & -d \\ 0 & -d & c \end{bmatrix} \begin{bmatrix} 0 & -b & 0 \\ 0 & 0 & a \\ 0 & a & -b \end{bmatrix}\right]$$

Phần tử DSG3

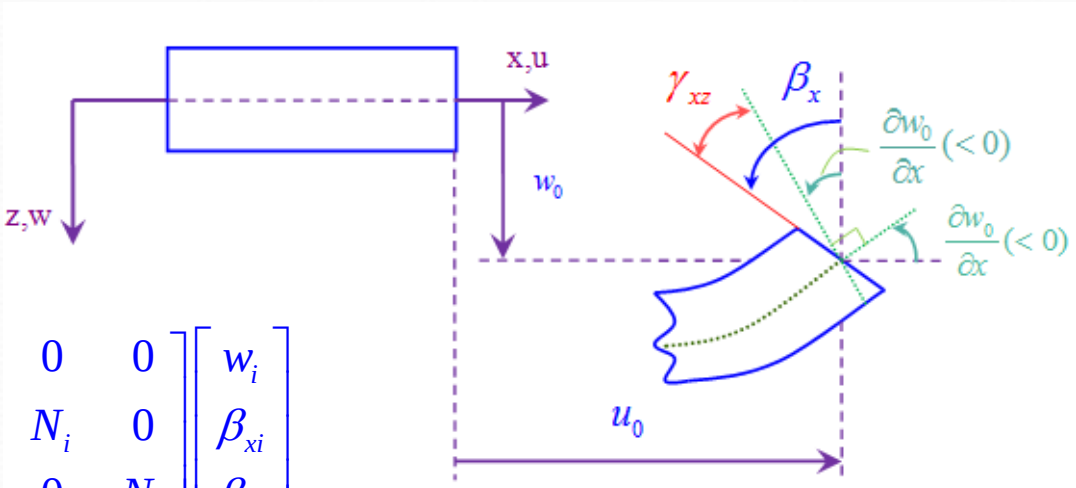
➢ Thành lập công thức Bs

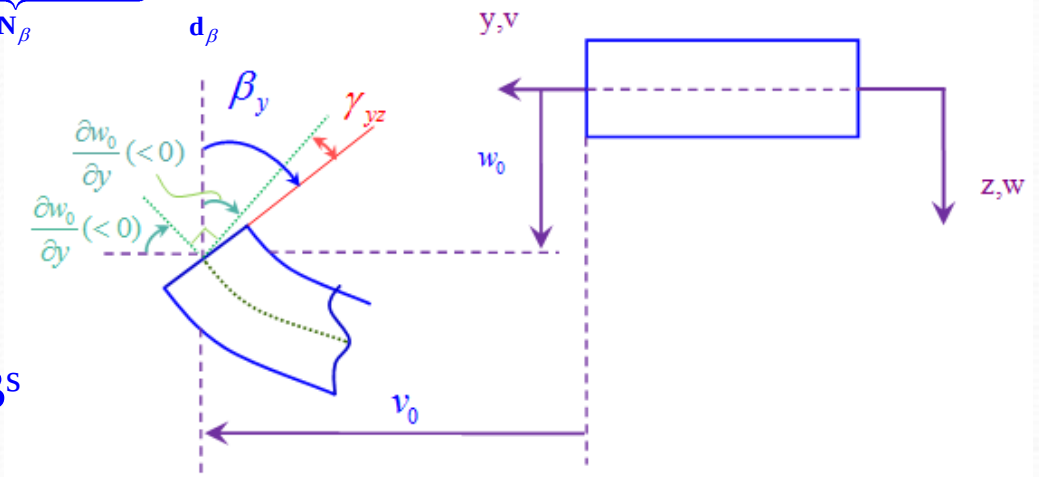
○ Biến dạng cắt

$$\gamma = \begin{bmatrix} \gamma_{xz} \\ \gamma_{yz} \end{bmatrix} = \begin{bmatrix} w'_x + \beta_x \\ w'_y + \beta_y \end{bmatrix}$$

$$\gamma = \begin{bmatrix} \partial_{,x} & 1 & 0 \\ \partial_{,y} & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix}}_{\mathbf{U}_\beta} = \begin{bmatrix} \partial_{,x} & 1 & 0 \\ \partial_{,y} & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta}$$

$$= \begin{bmatrix} N_{i,x} & N_i & 0 \\ N_{i,y} & 0 & N_i \end{bmatrix} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^s \mathbf{d}_\beta$$

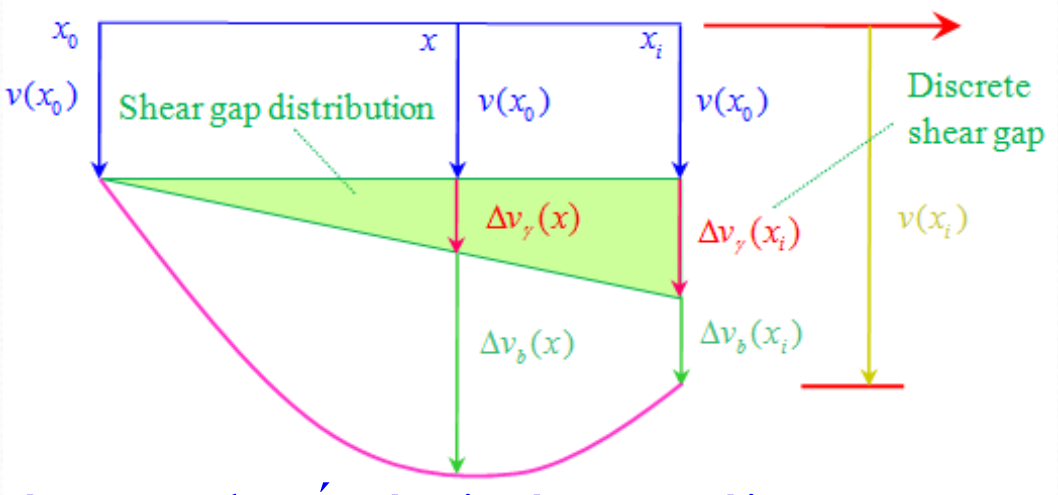




○ Shear locking: khác bậc của $\mathbf{B}^s$

Phần tử DSG3

- Khử shear locking trong ma trận **Bs**
  - Xét trường hợp một chiều 1D $\gamma(x) = w'(x) + \beta(x)$



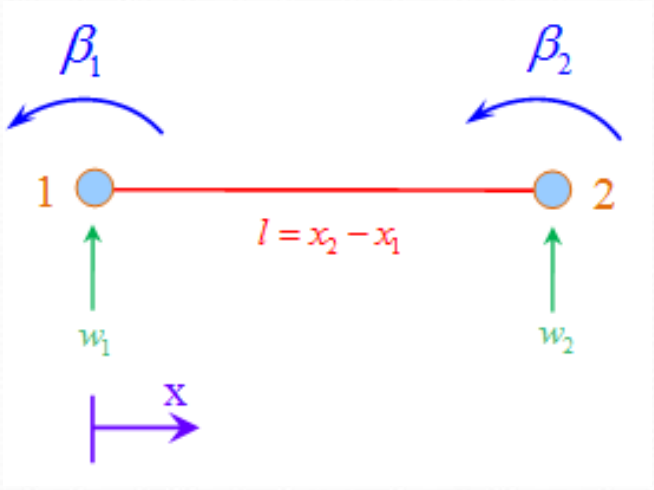
  - Độ lệch của góc xoay do cắt từ vị trí x_i so với x_0

$$\Delta v_\gamma(x_i) = \int_{x_0}^{x_i} \gamma(x)\mathbf{d}x = \int_{x_0}^{x_i} (w'(x) + \beta(x))\mathbf{d}x = w(x)\Big|_{x_0}^{x_i} + \int_{x_0}^{x_i} \beta(x)\mathbf{d}x$$

$$= \underbrace{w(x_i) - w(x_0)}_{\Delta v} + \Delta v_b$$

Phần tử DSG3

- Khử shear locking trong ma trận **B**s
  - Xét phần tử 2 node



-Tại vị trí node 1, đặt $\Delta v_\gamma^1 = 0$

-Xấp xỉ góc xoay $\beta(x) = \sum_{i=1}^{2} N_i \beta_i = \frac{l-x}{l}\beta_1 + \frac{x}{l}\beta_2$

-Tại vị trí node 2

$$\Delta v_\gamma^2(x_2) = \int_{x_1}^{x_2} \gamma(x)\mathbf{d}x = w(x)\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} \beta(x)\mathbf{d}x = w_2 - w_1 + \int_{x_1}^{x_2} \left(\frac{l-x}{l}\beta_1 + \frac{x}{l}\beta_2 \right) \mathbf{d}x$$

$$= w_2 - w_1 + \frac{1}{l}\left(\left(xl - \frac{x^2}{2} \right)_{x_1}^{x_2} \beta_1 + \left(\frac{x^2}{2} \right)_{x_1}^{x_2} \beta_2 \right) = w_2 - w_1 + \frac{l}{2}(\beta_1 + \beta_2)$$

Phần tử DSG3



- Khử shear locking trong ma trận **B**s
  - Có được giá trị độ lệch tại 2 node. Để khử shear locking (sự chênh lệch về bậc) ta xấp xỉ lại **1 trường tuyến tính của độ lệch** góc xoay do cắt

$$\Delta v_\gamma = \sum_{i=1}^{2} N_i \Delta v_\gamma^i = \frac{l-x}{l}\Delta v_\gamma^1 + \frac{x}{l}\Delta v_\gamma^2 = \frac{x}{l}\left[w_2 - w_1 + \frac{l}{2}\left(\beta_1 + \beta_2\right)\right]$$

0

  - Trường độ lệch góc xoay do cắt sau khi xấp xỉ lại đã không còn sự chệnh lệch về bậc ➔ khử được hiện tượng shear locking. Nhưng trường xấp xỉ mới chỉ trùng tại điểm đầu và điểm cuối so với trường cũ
  - Để tính tại mọi điểm

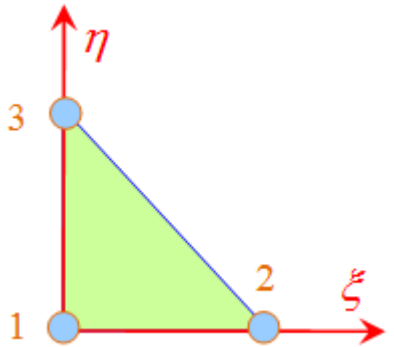
$$\gamma(x) = \frac{\mathbf{d}\Delta v_\gamma}{\mathbf{d}x} = \frac{1}{l}\left[w_2 - w_1 + \frac{l}{2}\left(\beta_1 + \beta_2\right)\right]$$

Phần tử DSG3

- Khử shear locking trong ma trận **B**s
  - Xét trường hợp một chiều cho mỗi phương (2D)

-Tính theo phương ξ ; $\Delta v_{\gamma 1}$

-Theo 1D $\Delta v_\gamma(\hat{x}) = w(x)\Big|_{x_0}^{\hat{x}} + \int_{x_0}^{\hat{x}} \beta(x)\mathbf{d}x$



$$\Rightarrow \Delta v_{\gamma 1} = w(\xi,\eta)\Big|_{\xi_1}^{\xi_i} + \int_{x_1}^{x_i} \beta_x \mathbf{d}x + \int_{y_1}^{y_i} \beta_y \mathbf{d}y$$

$$= w(\xi,\eta)\Big|_{\xi_1}^{\xi_i} + \int_{\xi_1}^{\xi_i} \beta_x \frac{\mathbf{d}x}{\mathbf{d}\xi}\mathbf{d}\xi + \int_{\xi_1}^{\xi_i} \beta_y \frac{\mathbf{d}y}{\mathbf{d}\xi}\mathbf{d}\xi$$

$$= w(\xi,\eta)\Big|_{\xi_1}^{\xi_i} + \int_{\xi_1}^{\xi_i} \left(\beta_x \mathrm{a} + \beta_y \mathrm{b}\right)\mathbf{d}\xi$$

$$\mathbf{J} = \begin{bmatrix} x_{,\xi} & y_{,\xi} \\ x_{,\eta} & y_{,\eta} \end{bmatrix} = \begin{bmatrix} \mathrm{a} & \mathrm{b} \\ \mathrm{d} & \mathrm{c} \end{bmatrix}$$

$$N_1 = 1-\xi-\eta \quad N_2 = \xi \quad N_3 = \eta$$

$$\mathbf{U}_\beta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}$$

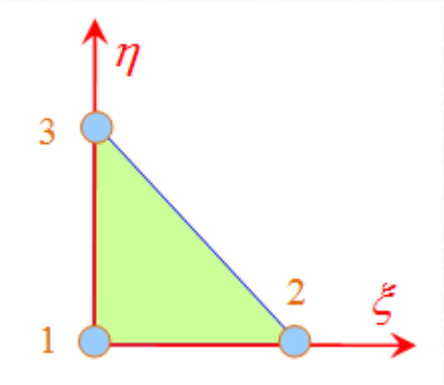
Phần tử DSG3

➢ Khử shear locking trong ma trận **Bs**

$$\Delta v_{\gamma 1} = w(\xi,\eta)\Big|_{\xi_1}^{\xi_i} + \int_{\xi_1}^{\xi_i} \left(\beta_x \mathrm{a} + \beta_y \mathrm{b}\right)\mathbf{d}\xi = w(\xi,\eta)\Big|_{\xi_1}^{\xi_i}$$

$$+ \int_{\xi_1}^{\xi_i} \mathrm{a}\left[\left(1-\xi-\eta\right)\beta_{x1} + \xi\beta_{x2} + \eta\beta_{x3}\right]\mathbf{d}\xi$$

$$+ \int_{\xi_1}^{\xi_i} \mathrm{b}\left[\left(1-\xi-\eta\right)\beta_{y1} + \xi\beta_{y2} + \eta\beta_{y3}\right]\mathbf{d}\xi$$



$$N_1 = 1-\xi-\eta \quad N_2 = \xi \quad N_3 = \eta$$

$$\Delta v_{\gamma 1} = w(\xi,\eta)\Big|_{\xi_1}^{\xi_i} + \mathrm{a}\left[\left(\xi - \frac{\xi^2}{2} - \eta\xi\right)\beta_{x1} + \frac{\xi^2}{2}\beta_{x2} + \eta\xi\beta_{x3}\right]_{\xi_1}^{\xi_i}$$

$$+ \mathrm{b}\left[\left(\xi - \frac{\xi^2}{2} - \eta\xi\right)\beta_{y1} + \frac{\xi^2}{2}\beta_{y2} + \eta\xi\beta_{y3}\right]_{\xi_1}^{\xi_i}$$

Phần tử DSG3

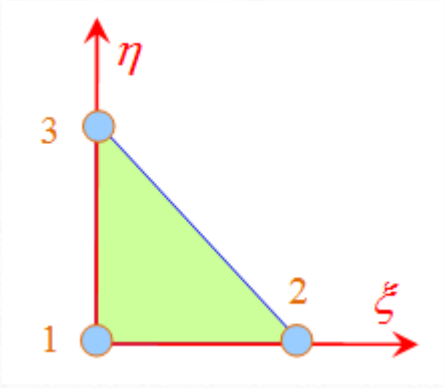
- Khử shear locking trong ma trận **B**s
  - Xét trường hợp một chiều 2D

-Tính theo phương η; $\Delta v_{\gamma 2}$

$$\Delta v_{\gamma 1} = w(\xi,\eta)\Big|_{\eta_1}^{\eta_i} + \int_{x_1}^{x_i} \beta_x \mathbf{d}x + \int_{y_1}^{y_i} \beta_y \mathbf{d}y$$

$$= w(\xi,\eta)\Big|_{\eta_1}^{\eta_i} + \int_{\eta_1}^{\eta_i} \beta_x \frac{\mathbf{d}x}{\mathbf{d}\eta}\mathbf{d}\eta + \int_{x_1}^{\eta_i} \beta_y \frac{\mathbf{d}y}{\mathbf{d}\eta}\mathbf{d}\eta$$

$$= w(\xi,\eta)\Big|_{\eta_1}^{\eta_i} + \int_{\eta_1}^{\eta_i} \left(\beta_x \mathrm{d} + \beta_y \mathrm{c}\right)\mathbf{d}\eta$$



$$\mathbf{J} = \begin{bmatrix} x_{,\xi} & y_{,\xi} \\ x_{,\eta} & y_{,\eta} \end{bmatrix} = \begin{bmatrix} \mathrm{a} & \mathrm{b} \\ \mathrm{d} & \mathrm{c} \end{bmatrix}$$

$$N_1 = 1-\xi-\eta \quad N_2 = \xi \quad N_3 = \eta$$

$$\mathbf{U}_\beta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}$$

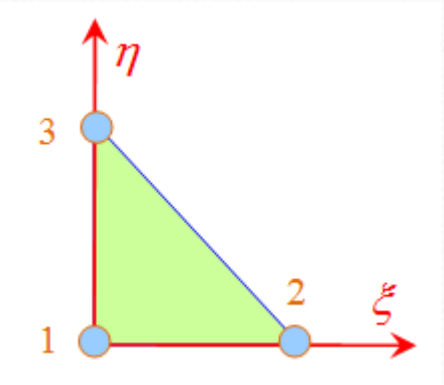
Phần tử DSG3

➢ Khử shear locking trong ma trận **Bs**

$$\Delta v_{\gamma 2} = w(\xi,\eta)\Big|_{\eta_1}^{\eta_i} + \int_{\eta_1}^{\eta_i} \left(\beta_x \mathrm{d} + \beta_y \mathrm{c}\right)\mathbf{d}\eta = w(\xi,\eta)\Big|_{\eta_1}^{\eta_i}$$

$$+ \int_{\eta_1}^{\eta_i} \mathrm{d}\left[\left(1-\xi-\eta\right)\beta_{x1} + \xi\beta_{x2} + \eta\beta_{x3}\right]\mathbf{d}\eta$$

$$+ \int_{\eta_1}^{\eta_i} \mathrm{c}\left[\left(1-\xi-\eta\right)\beta_{y1} + \xi\beta_{y2} + \eta\beta_{y3}\right]\mathbf{d}\eta$$



$$N_1 = 1-\xi-\eta \quad N_2 = \xi \quad N_3 = \eta$$

$$\Delta v_{\gamma 2} = w(\xi,\eta)\Big|_{\eta_1}^{\eta_i} + \mathrm{d}\left[\left(\eta - \frac{\eta^2}{2} - \eta\xi\right)\beta_{x1} + \frac{\eta^2}{2}\theta_{x3} + \eta\xi\beta_{x2}\right]_{\eta_1}^{\eta_i}$$

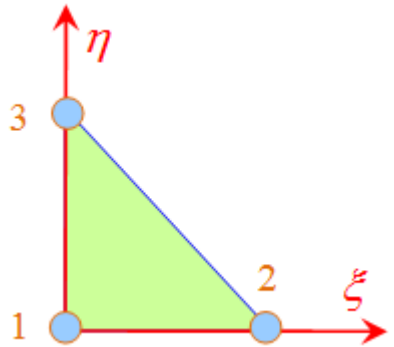
$$+ \mathrm{c}\left[\left(\eta - \frac{\eta^2}{2} - \eta\xi\right)\beta_{y1} + \frac{\eta^2}{2}\beta_{y3} + \eta\xi\beta_{y2}\right]_{\eta_1}^{\eta_i}$$

Phần tử DSG3

- Khử shear locking trong ma trận **Bs**

  -Tính cho từng node của phần tử

  - ✓ Node 1 $\Delta v_{\gamma 1}^{1}=0;\ \Delta v_{\gamma 2}^{1}=0$
  - ✓ Node 2: Phương ξ $\xi=1\ (\eta=0)$



$$\Delta v_{\gamma 1}^{2}=w(\xi,\eta)\Big|_{\xi_1}^{\xi_2}$$

$$+a\left[\left(\xi-\frac{\xi^2}{2}-\eta\xi\right)\beta_{x1}+\frac{\xi^2}{2}\beta_{x2}+\eta\xi\beta_{x3}\right]_{\xi_1}^{\xi_2}$$

$$+b\left[\left(\xi-\frac{\xi^2}{2}-\eta\xi\right)\beta_{y1}+\frac{\xi^2}{2}\beta_{y2}+\eta\xi\beta_{y3}\right]_{\xi_1}^{\xi_2}$$

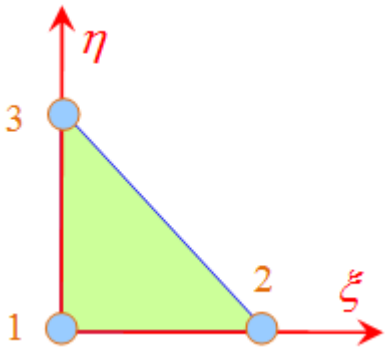
$$=w_2-w_1+\frac{a}{2}\left(\beta_{x1}+\beta_{x2}\right)+\frac{b}{2}\left(\beta_{y1}+\beta_{y2}\right)$$

-Phần tử DSG được thành lập trên cơ sở 1 chiều (1D) theo mỗi phương, nên theo mỗi phương chỉ có 1 giá trị

$$\Delta v_{\gamma 2}^{2}=w_2-w_1+\frac{a}{2}\left(\beta_{x1}+\beta_{x2}\right)+\frac{b}{2}\left(\beta_{y1}+\beta_{y2}\right)$$

Phần tử DSG3

- Khử shear locking trong ma trận **B**s
  - ✓ Node 3: Phương η $\eta = 1\ (\xi = 0)$



$$\Delta v_{\gamma 2}^{3} = w(\xi,\eta)\Big|_{\eta_1}^{\eta_3}$$

$$+ \mathrm{d}\left[\left(\eta - \frac{\eta^2}{2} - \eta\xi\right)\beta_{x1} + \frac{\eta^2}{2}\beta_{x2} + \eta\xi\beta_{x3}\right]_{\eta_1}^{\eta_3}$$

$$+ \mathrm{c}\left[\left(\eta - \frac{\eta^2}{2} - \eta\xi\right)\beta_{y1} + \frac{\eta^2}{2}\beta_{y2} + \eta\xi\beta_{y3}\right]_{\eta_1}^{\eta_3}$$

$$= w_3 - w_1 + \frac{\mathrm{d}}{2}(\beta_{x1} + \beta_{x2}) + \frac{\mathrm{c}}{2}(\beta_{y1} + \beta_{y2})$$

-Phần tử DSG được thành lập trên cơ sở 1 chiều (1D) theo mỗi phương, nên theo mỗi phương chỉ có 1 giá trị

$$\Delta v_{\gamma 1}^{3} = w_3 - w_1 + \frac{\mathrm{d}}{2}(\beta_{x1} + \beta_{x2}) + \frac{\mathrm{c}}{2}(\beta_{y1} + \beta_{y2})$$

Phần tử DSG3



➢ Khử shear locking trong ma trận **B**s

- Xấp xỉ lại trường tuyến tính của độ lệch góc xoay do cắt gây ra

$$\begin{bmatrix} \Delta\gamma_1 \\ \Delta\gamma_2 \end{bmatrix} = \sum_{i=1}^{3} N_i(\xi,\eta) \begin{bmatrix} \Delta v^i_{\gamma_1} \\ \Delta v^i_{\gamma_2} \end{bmatrix}$$

- Để tính tại mọi điểm

$$\begin{bmatrix} \gamma_x \\ \gamma_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial \Delta\gamma_1}{\partial x} \\ \dfrac{\partial \Delta\gamma_2}{\partial y} \end{bmatrix} = \begin{bmatrix} \displaystyle\sum_{i=1}^{3} \frac{\partial N_i}{\partial x} \Delta v^i_{\gamma_1} \\ \displaystyle\sum_{i=1}^{3} \frac{\partial N_i}{\partial y} \Delta v^i_{\gamma_2} \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{\partial N_1}{\partial x}\Delta v^1_{\gamma_1} + \dfrac{\partial N_2}{\partial x}\Delta v^2_{\gamma_1} + \dfrac{\partial N_3}{\partial x}\Delta v^3_{\gamma_1} \\ \dfrac{\partial N_1}{\partial y}\Delta v^1_{\gamma_2} + \dfrac{\partial N_2}{\partial y}\Delta v^2_{\gamma_2} + \dfrac{\partial N_3}{\partial y}\Delta v^3_{\gamma_2} \end{bmatrix}$$

Phần tử DSG3



➢ Khử shear locking trong ma trận **B**s

$$\begin{bmatrix}\gamma_x\\ \gamma_y\end{bmatrix}=\begin{bmatrix}\frac{\partial N_2}{\partial x}\Delta v^2_{\gamma_1}+\frac{\partial N_3}{\partial x}\Delta v^3_{\gamma_1}\\ \frac{\partial N_2}{\partial y}\Delta v^2_{\gamma_2}+\frac{\partial N_3}{\partial y}\Delta v^3_{\gamma_2}\end{bmatrix}=\begin{bmatrix}\frac{\partial N_2}{\partial \xi}\frac{\partial \xi}{\partial x}\Delta v^2_{\gamma_1}+\frac{\partial N_3}{\partial \eta}\frac{\partial \eta}{\partial x}\Delta v^3_{\gamma_1}\\ \frac{\partial N_2}{\partial \xi}\frac{\partial \xi}{\partial y}\Delta v^2_{\gamma_2}+\frac{\partial N_3}{\partial \eta}\frac{\partial \eta}{\partial y}\Delta v^3_{\gamma_2}\end{bmatrix}$$

✓ Node 2 $\Delta v^2_{\gamma 1}=w_2-w_1+\frac{a}{2}(\beta_{x1}+\beta_{x2})+\frac{b}{2}(\beta_{y1}+\beta_{y2})$

$\Delta v^2_{\gamma 1}=w_2-w_1+\frac{a}{2}(\beta_{x1}+\beta_{x2})+\frac{b}{2}(\beta_{y1}+\beta_{y2})$

so 2

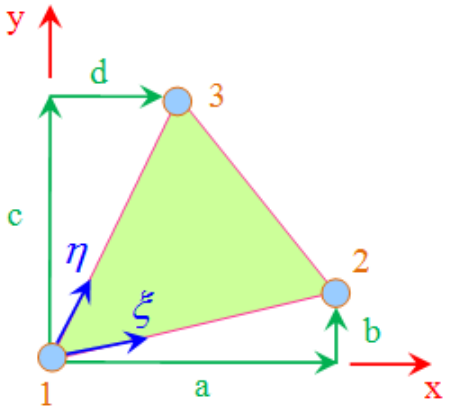
✓ Node 3 $\Delta v^3_{\gamma 1}=w_3-w_1+\frac{d}{2}(\beta_{x1}+\beta_{x2})+\frac{c}{2}(\beta_{y1}+\beta_{y2})$

$\Delta v^3_{\gamma 1}=w_3-w_1+\frac{d}{2}(\beta_{x1}+\beta_{x2})+\frac{c}{2}(\beta_{y1}+\beta_{y2})$

Phần tử DSG3

➢ Khử shear locking trong ma trận **B**s

$$\begin{bmatrix}\gamma_x\\ \gamma_y\end{bmatrix}=\begin{bmatrix}\dfrac{\partial N_2}{\partial\xi}\dfrac{\partial\xi}{\partial x}\Delta v^2_{\gamma_1}+\dfrac{\partial N_3}{\partial\eta}\dfrac{\partial\eta}{\partial x}\Delta v^3_{\gamma_1}\\ \dfrac{\partial N_2}{\partial\xi}\dfrac{\partial\xi}{\partial y}\Delta v^2_{\gamma_2}+\dfrac{\partial N_3}{\partial\eta}\dfrac{\partial\eta}{\partial y}\Delta v^3_{\gamma_2}\end{bmatrix}=\frac{1}{\det \mathbf{J}}\begin{bmatrix}c\Delta v^2_{\gamma_1}-b\Delta v^3_{\gamma_1}\\ -d\Delta v^2_{\gamma_2}+a\Delta v^3_{\gamma_2}\end{bmatrix}$$



$$\Delta v^2_{\gamma 1}=w_2-w_1+\frac{a}{2}(\beta_{x1}+\beta_{x2})+\frac{b}{2}(\beta_{y1}+\beta_{y2})$$

$$\Delta v^3_{\gamma 1}=w_3-w_1+\frac{d}{2}(\beta_{x1}+\beta_{x2})+\frac{c}{2}(\beta_{y1}+\beta_{y2})$$

$$N_{1,\xi}=-1 \quad N_{2,\xi}=1 \quad N_{3,\xi}=0$$

$$N_{1,\eta}=-1 \quad N_{2,\eta}=0 \quad N_{3,\eta}=1$$

$$\mathbf{J}^{-1}=\begin{bmatrix}\xi_{,x} & \eta_{,x}\\ \xi_{,y} & \eta_{,y}\end{bmatrix}=\frac{1}{\det \mathbf{J}}\begin{bmatrix}c & -b\\ -d & a\end{bmatrix}$$

$$\begin{bmatrix}\gamma_x\\ \gamma_y\end{bmatrix}=\frac{1}{\det \mathbf{J}}\begin{bmatrix}c\Delta v^2_{\gamma_1}-b\Delta v^3_{\gamma_1}\\ -d\Delta v^2_{\gamma_2}+a\Delta v^3_{\gamma_2}\end{bmatrix}$$

$$=\frac{1}{\det \mathbf{J}}\begin{bmatrix}(b-c)w_1+cw_2-bw_3+\frac{1}{2}\det \mathbf{J}\beta_{x1}+\frac{1}{2}ac\beta_{x2}-\frac{1}{2}bd\beta_{x3}+0\beta_{y1}+\frac{1}{2}bc\beta_{y2}-\frac{1}{2}bc\beta_{y3}\\ (d-a)w_1-dw_2+aw_3+0\beta_{x1}-\frac{1}{2}ad\beta_{x2}-\frac{1}{2}ad\beta_{x3}+\frac{1}{2}\det \mathbf{J}\beta_{y1}-\frac{1}{2}ab\beta_{y2}+\frac{1}{2}ac\beta_{y3}\end{bmatrix}$$

Phần tử DSG3



➢ Các công thức **B**

- Do uốn gây ra

$$\mathbf{B}^b = \frac{1}{\det \mathbf{J}} \begin{bmatrix} 0 & b-c & 0 \\ 0 & 0 & d-a \\ 0 & d-a & b-c \end{bmatrix} \begin{bmatrix} 0 & c & 0 \\ 0 & 0 & -d \\ 0 & -d & c \end{bmatrix} \begin{bmatrix} 0 & -b & 0 \\ 0 & 0 & a \\ 0 & a & -b \end{bmatrix}$$

- Do cắt gây ra

$$\mathbf{B}^s = \frac{1}{\det \mathbf{J}} \begin{bmatrix} b-c & \frac{\det \mathbf{J}}{2} & 0 \\ d-a & 0 & \frac{\det \mathbf{J}}{2} \end{bmatrix} \begin{bmatrix} c & \frac{ac}{2} & \frac{bc}{2} \\ -d & -\frac{ad}{2} & -\frac{bd}{2} \end{bmatrix} \begin{bmatrix} -b & -\frac{bd}{2} & -\frac{bc}{2} \\ a & \frac{ad}{2} & \frac{ac}{2} \end{bmatrix}$$

Phần tử DSG3

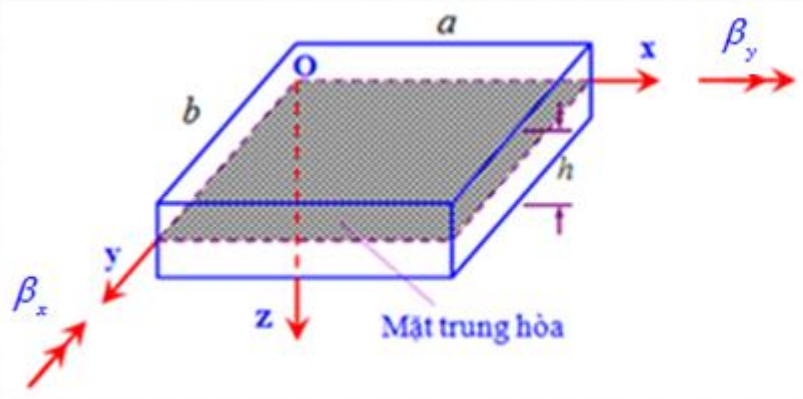
➢ Chuyển hệ trục tọa độ

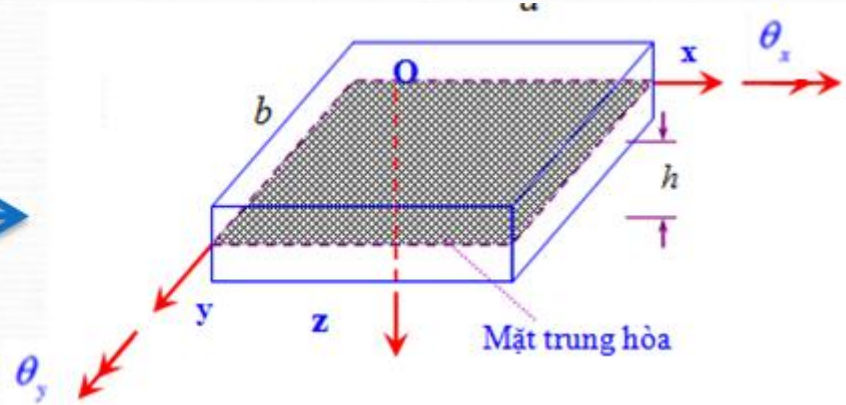
○ Để thuận tiện trong các phép biến đổi, lập trình, ta đổi biến

$$\beta_y = \theta_x \quad ; \quad \beta_x = -\theta_y$$

$$u = u_0 + z\beta_x$$
$$v = v_0 + z\beta_y$$
$$w = w_0$$

$$u = u_0 - z\theta_y$$
$$v = v_0 + z\theta_x$$
$$w = w_0$$





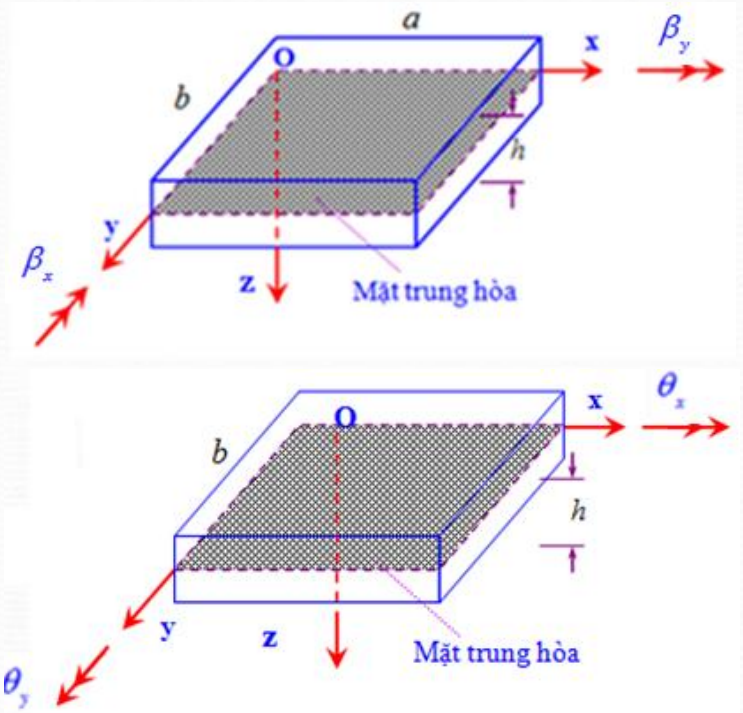
Hệ tọa độ chính ***Hệ tọa độ chuyển đổi***

Phần tử DSG3

➢ Chuyển hệ trục tọa độ

○ Biến xấp xỉ theo β $\quad \mathbf{U}_\beta = \mathbf{N}_\beta \mathbf{d}_\beta$

$$\mathbf{U}_\beta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} \begin{matrix} \Longrightarrow \text{Trục z} \\ \Longrightarrow \text{Trục y} \\ \Longrightarrow \text{Trục x} \end{matrix}$$



○ Biến xấp xỉ theo θ $\quad \mathbf{U}_\theta = \mathbf{N}_\theta \mathbf{d}_\theta$

$$\mathbf{U}_\theta = \begin{bmatrix} w \\ \theta_x \\ \theta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \theta_{xi} \\ \theta_{yi} \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \theta_{xi} \\ \theta_{yi} \end{bmatrix} \begin{matrix} \Longrightarrow \text{Trục z} \\ \Longrightarrow \text{Trục x} \\ \Longrightarrow \text{Trục y} \end{matrix}$$

Phần tử DSG3



➢ Chuyển hệ trục tọa độ

✓ Đổi biến $\beta_y = \theta_x \;;\; \beta_x = -\theta_y$

$$\mathbf{U}_\beta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} w \\ -\theta_y \\ \theta_x \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} \Leftrightarrow \mathbf{U}_\beta = \begin{bmatrix} w \\ \theta_y \\ \theta_x \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & -N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}$$

✓ Thay đổi vị trí của $\theta_x \,;\, \theta_y$

$$\underbrace{\begin{bmatrix} w \\ \theta_x \\ \theta_y \end{bmatrix}}_{\mathbf{U}_\theta} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & 0 & N_i \\ 0 & -N_i & 0 \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} = \underbrace{\underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}}_{\mathbf{T}}}_{\mathbf{N}_\theta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta}$$

Phần tử DSG3



- Chuyển hệ trục tọa độ
  - ✓ Tất cả các công thức đều được thiết lập trên hệ tọa độ chính

$$\kappa_\beta = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix}}_{\mathbf{U}_\beta} = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial y} \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} = \mathbf{B}^b \mathbf{d}_\beta$$

✓ Để thuận tiện lập trình và xuất kết quả nên ta chuyển qua hệ tọa độ chuyển đổi

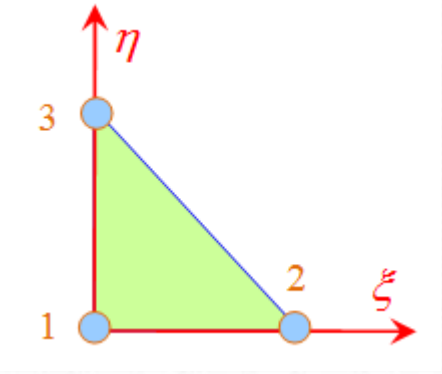
$$\kappa_\theta = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \mathbf{U}_\theta = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \mathbf{T} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial y} \end{bmatrix} \mathbf{T} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^b \mathbf{T} \mathbf{d}_\beta$$

Phần tử MIN3



➢ Công thức tính Bb

$$\mathbf{B}^{b}_{MIN3} = \frac{1}{\det \mathbf{J}} \begin{bmatrix} 0 & 0 & b_1 \\ 0 & a_1 & 0 \\ 0 & b_1 & a_1 \end{bmatrix} \begin{bmatrix} 0 & 0 & b_2 \\ 0 & a_2 & 0 \\ 0 & b_2 & a_2 \end{bmatrix} \begin{bmatrix} 0 & 0 & b_3 \\ 0 & a_3 & 0 \\ 0 & b_3 & a_3 \end{bmatrix}$$



$$\mathbf{B}^{b}_{DSG3} = \frac{1}{\det \mathbf{J}} \begin{bmatrix} 0 & b-c & 0 \\ 0 & 0 & d-a \\ 0 & d-a & b-c \end{bmatrix} \begin{bmatrix} 0 & c & 0 \\ 0 & 0 & -d \\ 0 & -d & c \end{bmatrix} \begin{bmatrix} 0 & -b & 0 \\ 0 & 0 & a \\ 0 & a & -b \end{bmatrix}$$

$$a_1 = x_3 - x_2 \qquad b_1 = y_2 - y_3$$

$$a_2 = x_1 - x_3 \qquad b_2 = y_3 - y_1$$

$$a_3 = x_2 - x_1 \qquad b_3 = y_1 - y_2$$

✓ Công thức **Bb** trong MIN3 được thành lập từ "hệ trục chuyển đổi".



Phần tử MIN3



➢ Khử shear locking trong Bs

✓ MIN3 khử shear bằng cách xấp xỉ lại trường tuyến tính của độ lệch thông qua 3 điểm Gauss nằm ở trung điểm của mỗi cạnh ➔ Trường Bs xấp xỉ lại chỉ trùng tại 3 điểm là trung điểm của mỗi cạnh

✓ DSG3 khử shear bằng cách xấp xỉ lại trường tuyến tính của độ lệch. Trường này được xấp xỉ là 1D theo mỗi phương tại các node ➔ Trường Bs xấp xỉ lại chỉ trùng tại 3 điểm là đỉnh của tam giác



Phần tử DKT



- Khử shear locking trong Bs
  - ✓ Trong DKT, khử shear locking giống như khử trong DSG3
- Công thức Bb
  - ✓ Trong DKT các trường được xấp xỉ thông qua tam giác 6 node.

	DSG3	MIN3	DKT
Bb	Xấp xỉ hàm dạng 3 node	Xấp xỉ hàm dạng 3 node	Xấp xỉ hàm dạng 6 node
Bs	Chỉ trùng tại 3 đỉnh của tam giác	Chỉ trùng tại 3 điểm giữa mỗi cạnh	Chỉ trùng tại 3 đỉnh của tam giác

THANKS YOUR ATTENTION

